

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SYMRISE AG and SYMRISE INC.,
Petitioner,

v.

INOLEX INVESTMENT CORPORATION,
Patent Owner.

IPR2023-01035
Patent 10,995,048 B2

Before ULRIKE W. JENKS, TINA E. HULSE, and
ROBERT A. POLLOCK, *Administrative Patent Judges*.

HULSE, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Granting-in-Part and Denying-in-Part
Patent Owner's Revised Motion to Amend
35 U.S.C. § 318(a)

I. INTRODUCTION

Symrise AG and Symrise Inc. (collectively, “Petitioner”) filed a Petition requesting an *inter partes* review of claims 1–6, 8–10, and 22–28 of U.S. Patent No. 10,995,048 B2 (Ex. 1001, “the ’048 Patent”), owned by Inolex Investment Corporation (“Patent Owner”). Paper 1 (“Pet.”). Upon considering the Petition and Patent Owner’s Preliminary Response (Paper 6), we instituted an *inter partes* review of the challenged claims of the ’048 Patent. Paper 7 (“Dec. Inst.” or “Institution Decision”).

After institution, Patent Owner filed a Non-Contingent Motion to Amend (Paper 12) and did not file a Patent Owner Response. Petitioner filed an Opposition to that Motion to Amend (Paper 14). We provided Preliminary Guidance in response to Patent Owner’s Motion to Amend. Paper 16 (“Prelim. Guid.”). Thereafter, Patent Owner filed a Revised Motion to Amend (Paper 18, “RMTA”), moving to cancel claims 1–6, 8–10, and 22–28 and substitute proposed claims 29–44. Petitioner filed an Opposition to the RMTA (Paper 20, “Pet. Opp. RMTA”), Patent Owner filed a Reply to the Opposition (Paper 25, “PO Reply RMTA”), and Petitioner filed a Sur-reply (Paper 29, “Pet. Sur-reply RMTA”).

An oral hearing was held on September 19, 2024. A transcript of the hearing has been entered in the record. Paper 32 (“Tr.”).

We have authority under 35 U.S.C. § 6. We issue this Final Written Decision under 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73. For the reasons that follow, we determine Petitioner has shown by a preponderance of the evidence that proposed substitute claims 29–44 are unpatentable as obvious. Therefore, Patent Owner’s Revised Motion to Amend is granted with respect to the cancellation of claims 1–6, 8–10, and 22–28, but denied with respect to proposed substitute claims 29–44.

A. Real Parties-in-Interest

Petitioner identifies itself as the real party-in-interest. Pet. 2–3.

Patent Owner identifies itself as the real party-in-interest. Paper 5, 1.

B. Related Proceedings

The parties both state that they are not aware of any judicial or administrative proceedings involving the '048 Patent. Pet. 3; Paper 5, 1.

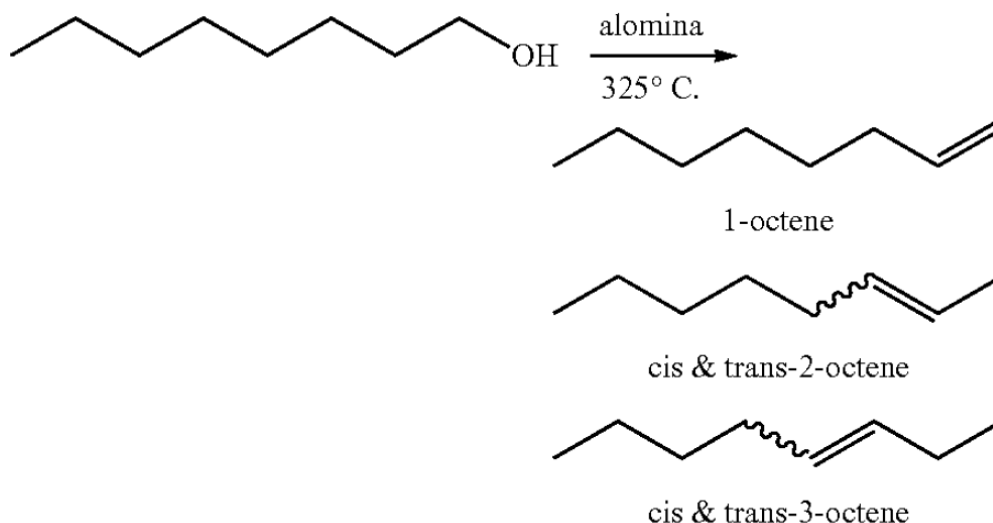
C. The '048 Patent

The '048 Patent, entitled “Natural 1,2-Alkanediols, Compositions Having Natural 1,2-Alkanediols and Processes for Making the Same,” was filed on November 13, 2019, and claims the benefit of U.S. Patent Application No. 16/262,902, filed on January 30, 2019, and of Provisional Application No. 62/623,958, filed on January 30, 2018. Ex. 1001, codes (54), (22), (63), (60).

The '048 Patent relates to natural and renewable antimicrobial compounds for use in alternative preservation systems for consumer products, such as personal care products. *Id.* at 1:25–31. Personal care products, like cosmetics, are packaged in clean but not sterile conditions. *Id.* at 1:39–42. As such, small amounts of microbes may be in products which, over time, may grow to a level significant enough to result in discoloration or a foul odor. *Id.* at 1:43–55. To address microbial growth, manufacturers add chemical preservatives, which are typically petrochemically based and derived from fossil fuels. *Id.* at 1:61–64. The Specification states that there is a desire to develop alternatives to traditional preservation and to use chemical components that are derived from sustainable and renewable resources (“feedstocks”). *Id.* at 2:13–16. Such “natural” and “renewable” products are formed by “new” carbon (versus “old” carbon from fossil fuels)

and are known in the art as non-petrochemically derived or “bio” products.
Id. at 3:35–40.

According to the Specification, caprylyl glycol (i.e., 1,2-octanediol) is one of the fastest growing alternative preservation components in the world.
Id. at 2:51–58. One known way to prepare caprylyl glycol is shown by the chemical reaction below:



Id. at 5:3–30.¹ The chemical reaction above shows the conversion of 1-octanol using heat and an acid catalyst (i.e., alumina) to form a mixture of 1-octene, cis- and trans-2-octene, and cis- and trans-3-octene.

Using a mixture of alkenes ultimately provides a mixture of diols upon dihydroxylation, providing a low yield of the desired terminal 1,2-diol material and producing unwanted contaminants. *Id.* at 5:7–10. But separating 2-, 3-, and 4-octene (not shown) regioisomers from the desired 1-octene is difficult and causes a lower net chemical yield of 1-octene. *Id.* at 6:2–5. Thus, the Specification states that “[t]here is a strong need in the art

¹ The chemical reaction appears to misspell alumina as “alomina.”

for a highly regioselective, and high-chemical conversion rate process for creating a bio-1-octene from a bio-1-octanol.” *Id.* at 6:22–24.

The Specification states that it is known that bio-1-butanol can convert to bio-1-butene with high selectivity and high chemical yield. *Id.* at 7:19–25. The ’048 Patent states, however, that “even with such a result for bio-1-butene, it is noted that the conversion of 1-octanol to 1-octene is not simply a homolog of a conversion of 1-butanol to bio-1-butene.” *Id.* at 7:39–42.

Accordingly, the Specification states that there is a need in the art for continued improvement to consumer products to provide additives to keep products safe from contamination while avoiding traditional preservatives and relying on components for consumer product compositions derived from natural bio-sources. *Id.* at 7:46–56. The Specification further states that “[t]here is more particularly a need for a natural, and preferably renewable/sustainable feedstock source to create a natural, bio-caprylyl glycol (i.e., a natural or bio-1,2-octanediol) and the associated bio-1-alkene intermediate.” *Id.* at 7:56–60.

D. Illustrative Issued Claim

The Petition challenges claims 1–6, 8–10, and 22–28 of the ’048 Patent, of which claim 1 is the only independent claim. Claim 1 is illustrative and reproduced below:

1. A method for making a bio-alkene, comprising:
dehydrating a bio-1-alcohol by heating the bio-1-alcohol in a reactor with a catalyst, wherein the resulting bio-alkene has a carbon chain of about 5 to about 20 carbon atoms and a bio-1-alkene regioselectivity of at least about 80%.

Ex. 1001, 29:45–50 (claim 1).

E. Illustrative Proposed Substitute Claim

In its Revised Motion to Amend, Patent Owner proposes canceling claims 1–6, 8–10, and 22–28 and substituting proposed substitute claims 29–44, respectively. RMTA 1, A-1. Proposed substitute claim 29 is reproduced below, showing the amendments to original claim 1 with the added language underlined and deleted language struck out:

29. A method for making a bio-alkene, comprising:

dehydrating a bio-1-alcohol by heating the bio-1-alcohol in a continuous flow reactor at a temperature of 240 °C to 360 °C with a calcium-promoted γ -alumina catalyst calcined at a temperature of about 420 °C to about 480 °C, wherein the catalyst has not been silanized, and

wherein the resulting bio-alkene has a carbon chain of 6 to 10 ~~about 5 to about 20~~ carbon atoms and is produced in a conversion of at least 95% % [sic] and has a bio-1-alkene regioselectivity of 95% to about 99% from a single-pass reaction after at least 20 hours time-on-stream-at least about 80%.

Id. at A-1.

F. The Asserted Grounds of Unpatentability

Because Patent Owner moves to cancel claims 1–6, 8–10, and 22–28, we analyze only the challenges to patentability of proposed substitute claims 29–44. Petitioner challenges the patentability of the proposed substitute claims on the following grounds:

Claims Challenged	35 U.S.C. §	Reference(s)/Basis
29–44	112	Written Description
29–44	112	Indefiniteness

Claims Challenged	35 U.S.C. §	Reference(s)/Basis
29–44	103	Wright '451 ²
29–44	103	De Bruyn, ³ Takamura ⁴
29–44	N/A	Estoppel under 37 C.F.R. § 42.73

Pet. Opp. RMTA 1.

Petitioner also relies on the Declarations of Douglass F. Taber, Ph.D. Exs. 1002, 1038, 1046. And Patent Owner relies on the Declarations of Dr. Michael E. Wright, Ph.D. Exs. 2001, 2006, 2010.

II. REVISED MOTION TO AMEND ANALYSIS

In an *inter partes* review, amended claims are not added to a patent as of right, but rather must be proposed as part of a motion to amend. 35 U.S.C. § 316(d). In its Non-Contingent Revised Motion to Amend, Patent Owner moves to cancel claims 1–6, 8–10, and 22–28 of the '048 Patent and proposes to substitute claims 29–44. RMTA 1, A-1. Petitioner opposes the addition of the substitute claims, asserting, among other reasons, that the proposed claims are unpatentable for lack of written description support, for indefiniteness, and as obvious. Pet. Opp. RMTA 1.

The Board must assess the patentability of proposed substitute claims “without placing the burden of persuasion on the patent owner.” *Aqua Prods., Inc. v. Matal*, 872 F.3d 1290, 1328 (Fed. Cir. 2017) (en banc); see

² Wright et al., WO 2015/130451 A1, published Sept. 3, 2015. Ex. 1039 (“Wright '451”).

³ De Bruyn et al., US 2005/0065389 A1, published Mar. 24, 2005. Ex. 1008 (“De Bruyn”).

⁴ Takamura et al., EP 2 990 395 A1, published Mar. 2, 2016. Ex. 1044 (“Takamura”).

also Lectrosonics, Inc. v. Zaxcom, Inc., IPR2018-01129, Paper 15 at 3–4 (PTAB Feb. 25, 2019) (precedential). The petitioner must prove that the proposed substitute claims are unpatentable by a preponderance of the evidence. *Bosch Auto. Serv. Sols., LLC v. Matal*, 878 F.3d 1027, 1040 (Fed. Cir. 2017); *Lectrosonics*, Paper 15 at 3–4.

As explained further below, having considered the arguments and evidence presented by the parties at trial, we determine Petitioner has shown by a preponderance of the evidence that proposed substitute claims 29–44 are unpatentable as obvious.

A. Person of Ordinary Skill in the Art

In determining the level of ordinary skill in the art, we consider the type of problems encountered in the art, the prior art solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the educational level of active workers in the field. *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986).

Petitioner contends that a person of ordinary skill in the art at the time of the invention (“POSA” or “POSITA”) would have had “a bachelor’s degree and two or more years of experience or a master’s degree in organic synthesis or chemical engineering, with education substituting for experience and *vice versa*.” Pet. 15 (citing Ex. 1002 ¶ 16). Patent Owner took no position on the level of ordinary skill in the art in its Preliminary Response. Prelim. Resp. 8.

In our Institution Decision, we adopted Petitioner’s definition. Dec. Inst. 6–7. In its Motion to Amend, Patent Owner states that it accepts the Board’s definition for purposes of this proceeding. MTA 24. We, therefore, see no reason to disturb our definition of the level of ordinary skill in the art

as it appears to be consistent with the prior art. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) (explaining that specific findings regarding ordinary skill level are not required “where the prior art itself reflects an appropriate level and a need for testimony is not shown” (quoting *Litton Indus. Prods., Inc. v. Solid State Sys. Corp.*, 755 F.2d 158, 163 (Fed. Cir. 1985))).

Moreover, we have reviewed the credentials of Dr. Taber and Dr. Wright and find they are both qualified to opine from the perspective of a skilled artisan as they both possess at least ordinary skill in the art.⁵ *See* Ex. 1002 ¶¶ 2–4, App’x A; Ex. 2001 ¶¶ 3–7, App’x A; *see also Kyocera Senco Indus. Tools Inc. v. Int’l Trade Comm’n*, 22 F.4th 1369, 1376–77 (Fed. Cir. 2022) (“To offer expert testimony from the perspective of a skilled artisan in a patent case . . . a witness must at least have ordinary skill in the art.”).

B. Claim Construction

In an *inter partes* review, the Board applies the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b). 37 C.F.R. § 100(b). Under that standard, claim terms “are generally given their ordinary and customary meaning” as understood by a person of ordinary skill in the art at the time of the invention. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

⁵ Although Dr. Wright questions Dr. Taber’s qualifications and whether he can “operate as [a] POSITA, let alone be considered an expert, in the technology of the ’048 patent,” Ex. 2006 ¶ 7, Patent Owner does not challenge Dr. Taber’s qualifications as an expert in any of its papers, and conceded during oral argument that Dr. Taber “certainly is a person of ordinary skill in the art.” Tr. 25:16–26.

Patent Owner proposes constructions for the terms “about,” “a calcium-promoted γ -alumina catalyst,” and “from a single-pass reaction.” RMTA 24–25. Petitioner does not contest the construction of any of those terms. *See generally* Pet. Opp. RMTA.

Although neither party expressly proposes a construction for the term “conversion,” the parties appear to differ in their interpretations of that term. Petitioner interprets “conversion” to mean the percent conversion of the 1-alcohol. *See* Pet. Opp. RMTA 6–7 (interpreting “conversion” as percentage of remaining 1-alcohol); Ex. 1040, 109:2–11. Patent Owner, on the other hand, interprets “conversion” to mean the total percentage of alkenes that have been produced. Ex. 2010 ¶ 123 (interpreting “conversion” as percentage of total alkenes). During oral argument, counsel for Patent Owner conceded that “conversion can mean either.” Tr. 27:18–28:13.

Without the benefit of briefing by the parties, we decline to expressly construe the term “conversion,” as our Decision would be the same under either interpretation, as discussed further below. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (finding only those terms that are in controversy need be construed, and only to the extent necessary to resolve the controversy). Moreover, we do not find it necessary to expressly construe any other claim terms for purposes of this Decision. *Id.*

C. Cancellation of Claims 1–6, 8–10, and 22–28

Patent Owner’s Revised Motion to Amend requests that claims 1–6, 8–10, and 22–28 be canceled. RMTA 1. The RMTA is not contingent on any finding by the Board regarding the patentability of claims 1–6, 8–10, and 22–28. *Id.* Rather, the RMTA seeks to replace challenged claims 1–6,

8–10, and 22–28 with proposed substitute claims 29–44 without precondition. *Id.*

A motion to amend may cancel a challenged claim. *See* 35 U.S.C. § 316(d)(1)(A); 37 C.F.R. § 42.121(a)(3). Petitioner does not oppose the cancellation of claims 1–6, 8–10, and 22–28. Accordingly, we grant Patent Owner’s Revised Motion to Amend with respect to the request to cancel challenged claims 1–6, 8–10, and 22–28.

D. Asserted Obviousness of Proposed Substitute Claims 29–44 over Wright ’451

Patent Owner seeks to add proposed substitute claims 29–44 in its Revised Motion to Amend. Petitioner asserts that substitute claims 29–44 are unpatentable as obvious over Wright ’451. Pet. Opp. RMTA 5–15.

Having considered the arguments and evidence presented by the parties and the record as a whole, we determine Petitioner has established by a preponderance of the evidence that substitute claims 29–44 are unpatentable as obvious over Wright ’451.⁶

1. Wright ’451

Wright ’451 is a PCT application entitled “Regioselective Dehydration of Terminal Alcohols” that published on September 3, 2015, which is more than one year before the earliest effective filing date of the ’048 Patent on January 30, 2018. *See* Ex. 1039, codes (54), (43); Ex. 1001, codes (60), (63). Wright ’451 is, therefore, prior art under 35 U.S.C.

⁶ As set forth below, we determine Petitioner has shown by a preponderance of the evidence that substitute claims 29–44 are unpatentable as obvious over Wright ’451. In light of that determination, we need not address whether Patent Owner’s Revised Motion to Amend otherwise complies with the statutory and regulatory requirements for a motion to amend. *See* 35 U.S.C. § 316(d); 37 C.F.R. § 42.121(a).

§ 102(b) and Patent Owner does not argue otherwise. Of note, the lead inventor of Wright '451, Michael E. Wright, is also an inventor of the '048 Patent and is Patent Owner's expert in this proceeding. *Compare id.*, code (72) with Ex. 1001, code (72).

Wright '451 "relates to conversion of terminal alcohols to the corresponding terminal alkenes with high regioselectivity at high conversion." *Id.* ¶ 2. Wright '451 states that for α -olefins (i.e., 1-alkenes) from petroleum resources, "there exist several well-known processes for ethylene oligomerization that afford α -olefins such as 1-butene, 1-hexene, 1-octene, and so forth." *Id.* ¶ 3. According to Wright '451, however, obtaining α -olefins from renewable and sustainable resources requires a "significantly different approach." *Id.* ¶ 4. Wright '451 explains that because fermentation can produce alcohols in large scale, alcohols can be desirable feedstock for preparing α -olefins, as long as they can be dehydrated in high yield, which requires achieving high regioselectivity at high alcohol conversion. *Id.* According to Wright '451, the problem with selective α -olefin synthesis is the competing isomerization reaction that produces the thermodynamically favored internal olefins. *Id.* Moreover, bio-alcohol fermentation processing directly results in water and impurities, and removing them is costly in energy and time. *Id.* ¶ 6. Accordingly, Wright '451 states "[a]n improved method is needed to prepare terminal olefins efficiently from a solution containing bio-1-alcohols and water as a major impurity." *Id.*

Wright '451 teaches a catalyst system and process that can convert a feed of terminal alcohol-containing water to a product stream containing the respective terminal olefin and water. *Id.* ¶ 13. In some embodiments, Wright '451 states "the terminal olefin is produced with about 90% to about

99% regiochemical selectivity,” and “a single pass over the catalyst system affords a chemical conversion of greater than about 95%.” *Id.* Moreover, Wright ’451 teaches that the catalyst may be calcined at temperatures of about 200 °C to 600 °C. *Id.* ¶ 67.

Wright ’451 teaches the modification of a catalyst composition, such as γ -alumina, with a calcium compound and calcining the catalyst composition. *Id.* ¶¶ 13–14. Wright ’451 states that the alcohol may be a bio-1-alcohol and may be “of general formula $R^1R^2CHCH_2OH$, where R^1 and R^2 are independently H or a C_1 to C_{20} alkyl radical.” *Id.* ¶¶ 20, 78. Wright ’451 further describes its catalyst system as providing “highly specific regioselective dehydration of 1-alcohols (*e.g.*, bio-1-alcohols) in water-containing mixtures, with retention of selectivity over extended periods of use, commonly reported as Time On Sream (TOS),” such as continuous uses “for at least 1 to 3 months.” *Id.* ¶ 50.

2. Analysis

A patent claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which said claimed invention pertains. *See also KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). Objective evidence of nonobviousness may include

“commercial success, long felt but unsolved needs, failure of others, etc.”

Id.

a) Teachings of Wright '451

Petitioner argues that Wright '451 teaches or suggests each limitation of substitute claims 29–44. Pet. Opp. RMTA 12–15. We agree for the reasons stated by Petitioner, which we adopt here. *Id.* (citing, e.g., Ex. 1039 ¶¶ 2, 13, 14, 16, 17, 50, 67, 78; Ex. 1046 ¶¶ 63–69). For example, regarding substitute claim 29, we find Wright '451 teaches methods for dehydrating bio-1-alcohols to the corresponding bio-1-alkenes using calcined and calcium-modified γ -alumina catalysts in a continuous flow reactor. Ex. 1039 ¶¶ 2, 14–18, 20. Wright '451 teaches that the catalyst, which has not been silanized,⁷ is calcined at 200 °C to 600 °C, and the dehydration reaction is between 220 °C to 440 °C. *Id.* ¶¶ 13, 67. Wright '451 also teaches that in some embodiments, “the regioselectivity of 1-alkene production is greater than about 95%.” *Id.* ¶ 22; see also *id.* ¶ 13 (“[T]he terminal olefin is produced with about 90% to about 99% regiochemical selectivity.”). Moreover, Wright '451 teaches that “a single pass over the catalyst system affords a chemical conversion of greater than about 95%.”⁸ *Id.* ¶ 13. And Wright '451 teaches that the Time On Stream can be “up to 48 months while producing terminal olefin (*e.g.*, bio-1-olefin) without

⁷ As explained in our Preliminary Guidance—which neither party disputes—we interpret “catalyst has not been silanized” to mean that no affirmative step is taken to silanize the catalyst. Prelim. Guid. 8.

⁸ We note that although Wright '451 does not indicate whether the cited “conversion” is calculated as a percentage of remaining 1-alcohol (as Petitioner would interpret it) or as the percentage of total alkenes (as Patent Owner would interpret it), Patent Owner does not contest that Wright '451 teaches the “conversion of at least 95%” limitation of the proposed substituted claims.

significant loss in regioselectivity for the terminal olefin.” *Id.* ¶ 50. Thus, we find a preponderance of the evidence shows that Wright ’451 teaches or suggests each limitation of claim 29.

We have considered the arguments and evidence presented by the parties with respect to substitute claims 30–44 and find a preponderance of the evidence shows that Wright ’451 teaches or suggestions each limitation of those claims, as well. *See, e.g.*, Ex. 1039 ¶¶ 14, 67, 78, 82; Ex. 1046 ¶¶ 67–72.

In response, Patent Owner argues that the level of ordinary skill in the art is low and that Dr. Taber “fail[s] to analyze the prior art through this lens of a POSITA being one who had little practical experience.” PO Reply RMTA 10. Citing Dr. Wright’s testimony, Patent Owner contends that “a POSITA would not have found the claimed calcination temperature ranges for a calcium-promoted γ -alumina catalyst of ‘about 420 °C to about 480 °C’ and ‘about 440 °C to about 460 °C’ are disclosed or suggested by the prior art.” *Id.* at 10–11 (citing Ex. 2010 ¶¶ 111–116⁹). Dr. Wright opines that Wright ’451’s disclosure of calcining at about 200°C to 600°C does not disclose or suggest the claimed calcination ranges because “[a] POSITA reading *Wright ’451* would have no reason to lower the calcination temperature below 500 °C because the catalysts already provide high conversion, high regioselectivity and long TOS for 1-butanol.” Ex. 2010 ¶ 114. Dr. Wright also challenges the credibility of Dr. Taber’s testimony, questioning his knowledge and conclusions. *Id.* ¶ 113.

⁹ Petitioner objects to Patent Owner’s argument, asserting that Patent Owner improperly incorporates Dr. Wright’s cited testimony by reference. Paper 27. We agree with Petitioner’s objection, but we are not persuaded by Patent Owner’s argument regardless.

We are not persuaded by Patent Owner’s arguments. This is a classic obviousness case of overlapping ranges with longstanding caselaw that holds: “Where a claimed range overlaps with a range disclosed in the prior art, there is a presumption of obviousness.” *Ormco Corp. v. Align Tech.*, 463 F.3d 1299, 1311 (Fed. Cir. 2006); *see also In re Peterson*, 315 F.3d 1325, 1329 (Fed. Cir. 2003) (citing *In re Geisler*, 116 F.3d 1465, 1469 (Fed. Cir. 1997); *In re Woodruff*, 919 F.2d 1575, 1578 (CCPA 1990); *In re Malagari*, 499 F.2d 1297, 1303 (CCPA 1974)). Wright ’451 discloses a reaction temperature of about 220 °C to about 440 °C (Ex. 1039 ¶ 13) and a calcination temperature range of 200 °C to 600 °C (*id.* ¶ 67), which both fully encompass the claimed reaction temperature range of “240 °C to 360 °C” and calcination temperature range of “about 420 °C to about 480 °C” of substitute claims 29–44. RMTA A-1.

As the Federal Circuit has explained, “[t]he point of our overlapping range cases is that, in the absence of evidence indicating that there is something special or critical about the claimed range, an overlap suffices to show that the claimed range was disclosed in—and therefore obvious in light of—the prior art.” *E.I. DuPont de Nemours & Co. v. Synvina C.V.*, 904 F.3d 996, 1008 (Fed. Cir. 2018). Thus, Petitioner need not specifically assert that a POSA would have been motivated “to lower the calcination temperature below 500 °C,” as Patent Owner and Dr. Wright assert. *See* Ex. 2010 ¶ 114. Showing that Wright ’451’s temperature ranges encompass the claimed ranges is sufficient to establish a presumption of obviousness. *See In re Peterson*, 315 F.3d at 1330 (concluding that a prior art reference that discloses a range fully encompassing a narrower claimed range is sufficient to establish a prima facie case of obviousness). Consistent with the Federal Circuit, we find that “[t]he normal desire of scientists or artisans to improve

upon what is already generally known provides the motivation to determine where in a disclosed set of percentage ranges is the optimum combination of percentages.” *Id.*

Having found that Wright ’451 teaches or suggests each limitation of the substitute claims and established that the claimed temperature ranges are presumed obvious, “the burden of production falls upon the patentee to come forward with evidence of teaching away, unexpected results or criticality, or other pertinent objective indicia indicating that the overlapping range would not have been obvious in light of that prior art.” *See E.I. DuPont*, 904 F.3d at 1008 (holding the burden shifting framework of overlapping ranges applies in *inter partes* review proceedings).

Accordingly, we now turn to the evidence of nonobviousness that Patent Owner has presented and address whether that evidence is sufficient to rebut the presumption that the proposed substitute claims are obvious. For the reasons set forth below, we find it is not.

b) Objective Evidence of Nonobviousness

(1) The Parties’ Contentions

Patent Owner relies primarily on evidence of unexpected results to rebut the presumption of obviousness over the overlapping temperature ranges of Wright ’451. RMTA 18–21. Specifically, Patent Owner asserts that (1) the calcination temperatures of substitute claims 29–44 provide unexpected results, and (2) the reaction temperatures of substitute claims 29–44 provide unexpected results. *Id.* at 19–21.

Regarding calcination temperature, Patent Owner argues that “[a]s shown in the ’048 patent, a calcium-promoted γ -alumina catalyst calcined at 440 °C provides unexpectedly better 1-octene regioselectivity and higher conversion compared to the same catalyst calcined at 500 °C.” *Id.* at 19

(citing Ex. 1001, 26:10–27:9 (Examples 1–3); Ex. 2006 ¶¶ 20, 123). Patent Owner cites Examples 1–3 of the '048 Patent in which Example 1 prepares a catalyst that is calcined at 440 °C and Example 2 prepares a catalyst that is calcined at 500°C. Ex. 1001, 26:40–43, 26:47–50. Example 3 uses the catalysts of Examples 1 and 2 to convert bio-1-octanol to bio-1-octene. *Id.* at 26:55–27:9. The results are shown in Table 2, which is reproduced below:

TABLE 2

Catalyst	Calcination Temperature (° C.)	1-Octene (%)	2-Octene & 3-Octenes (%)	1-Octanol (%)	1-Octene Selectivity (%)
EXAMPLE 1	440	94.0	3.2	1.6	97.4
EXAMPLE 2	500	88.9	6.6	3.5	93.1

Id. at 27:1–9. According to Patent Owner, Table 2 shows that “the calcium-promoted γ -alumina catalyst calcined at 500 °C provided a 1-octene regioselectivity of 93.1% at a conversion of 95.5%,¹⁰ while calcination at 440 °C provided a 1-octene regioselectivity of 96.7%¹¹ at a conversion of 97.2%.” RMTA 19 (citing Ex. 1001, 26:10–27:9; Ex. 2006 ¶¶ 20, 123). According to Patent Owner, “the '048 patent demonstrates that the calcination temperature of 500 °C (as taught by *Wright '451*) provides a *less reactive and less selective* catalyst than that of the present invention.” *Id.*

¹⁰ As explained above, Patent Owner and Dr. Wright calculate “conversion” of the “resulting bio-alkene” by adding the percentage of 1-Octene and the percentage of 2-Octene & 3-Octenes. Tr. 26:24–28:13; *see also* Ex. 2006 ¶ 123 (stating conversion (to octenes) in Table 2 is 97.2% (i.e., 94.0% + 3.2%) and 95.5% (i.e., 88.9% + 6.6%)).

¹¹ According to Dr. Wright, the calculation of 1-Octene selectivity for Example 1 “appears to be incorrect and should be 96.7%.” Ex. 2006 ¶ 123 n.4.

(citing Ex. 2006 ¶ 123). Patent Owner contends that one of ordinary skill in the art would have viewed this result as unexpected because “nothing in *Wright* ’451 (or any other prior art of record) teaches or suggests lower calcination temperature can improve both conversion and 1-alkene regioselectivity.” *Id.* (citing Ex. 2006 ¶ 124).

Patent Owner also argues that “this result is a difference in kind and not one of degree, as even a 1-2% improvement in either conversion or 1-alkene regioselectivity can provide significant benefit from the perspective of reactor design and operating costs—for example, by eliminating a recycle stream.” *Id.* (citing Ex. 2006 ¶¶ 6, 20, 122–124). Specifically, Patent Owner and Dr. Wright assert that “1-octene regioselectivity increased from 93.1% to 96.7% and conversion increased from 95.5% to 97.2%” and one of ordinary skill in the art would view these improvements as “very significant and highly unexpected.” *Id.* at 19–20 (citing Ex. 2006 ¶¶ 21, 123–124).

Regarding reaction temperature, Patent Owner relies on the results shown in Table 5, reproduced below:

TABLE 5

Catalyst	Reactor Temperature (° C.)	1-Hexene (%)	2-Hexenes & 3-Hexene (%)	1-Hexanol (%)	1-Hexene Selectivity (%)
EXAMPLE 11A	330	92.8	3.5	1.8	95.8
EXAMPLE 11B	350	93.8	3.1	1.7	96.8
EXAMPLE 11C	370	90.1	3.2	1.6	96.6

As shown in Table 5, Patent Owner asserts that “Example 11 of the ’048 [Patent] discloses that increasing the reactor temperature from 350 °C to 370 °C reduced conversion (to hexenes) from 96.9% to 93.3%, while 1-hexene regioselectivity was constant (96.6% vs. 96.4%).” *Id.* at 20 (citing Ex. 1001, 29:1–25; Ex. 2006 ¶¶ 126–127).

Patent Owner argues the decreased conversion of 1-hexanol at a reaction temperature of 370 °C relative to the conversion at 350 °C is an unexpected result. *Id.* (citing Ex. 2006 ¶¶ 126–127). According to Patent Owner, “*Wright ’451* teaches that the lowest 1-butanol conversion is always at 340 °C and *conversion increases with temperature*” and “[n]othing in *Wright ’451* would have taught or suggested to a POSITA that conversion of 1-hexanol would decrease at 370 °C.” *Id.* (citing Ex. 1039 ¶¶ 42–46; Ex. 2006 ¶¶ 125–127). Patent Owner again argues that the asserted unexpected result is a difference in kind, not one of degree, because “even a 1-2% improvement in conversion can provide significant design and operational benefits.” *Id.* (citing Ex. 2006 ¶¶ 125–127). Patent Owner also challenges the credibility of Dr. Taber’s testimony, stating his testimony “concerning a POSITA’s understanding of the operation and mechanism of the dehydration process at issue is not credible and therefore any testimony he attempts to provide in rebuttal concerning whether a result would be expected or unexpected should be accorded little probative value.” *Id.* at 21 (citing Ex. 2006 ¶¶ 7–14, 63, 73, 87, 89, 99, 114, 115).

In response, Petitioner asserts that “[i]t is undisputed that *Wright ’451* teaches dehydrating a 1-alcohol at 240 °C to 360 °C in a single pass using a catalyst calcined at 200 °C to 600 °C that achieves 95-99% regioselectivity and at least 95% conversion.” Pet. Opp. RMTA 5 (citing Ex. 1039 ¶¶ 13–14, 20, 22, 67, 78, 82; RMTA 18–20; Ex. 2006 ¶¶ 66–70; Ex. 1046 ¶ 43). Petitioner disagrees that the claimed reaction and calcination temperatures produce unexpected conversion and regioselectivity results. *Id.* at 6–7 (citing Ex. 1046 ¶¶ 45–61). Petitioner asserts that Patent Owner and Dr. Wright misinterpret the experiment set forth in Example 11 and the results in Table 5. *Id.* at 6. Specifically, Petitioner interprets “conversion”

to mean the conversion of 1-hexanol and asserts that Patent Owner's interpretation (i.e., the sum of the resulting alkenes) is incorrect. *Id.* at 6–7. Nevertheless, Petitioner argues that the results are consistent with the prior art teachings and are not unexpected. *Id.* Petitioner also argues that neither the prosecution history nor the specification suggest the claimed ranges are critical. *Id.* at 10–11.

In its Reply to Petitioner's Opposition, Patent Owner argues that Petitioner has failed to rebut Patent Owner's asserted unexpected results, reiterating its position that the '048 Patent's Example 3 shows unexpected results when calcined at 440 °C as compared to 500 °C. PO Reply RMTA 11–12 (citing Ex. 2010 ¶¶ 114–121, 133, 134). Patent Owner also argues that a POSA would have found it unexpected that the '048 Patent's unsilanized calcium-promoted, γ -alumina catalyst would provide both increased conversion and regioselectivity compared to prior art catalysts that were base-treated, dried, silanized, and then calcined at 400 °C. *Id.* at 12 (citing Ex. 2010 ¶ 122). Patent Owner also argues that a POSA would not have expected that increasing the reactor temperature would reduce conversion of 1-hexanol to 1-hexene, while maintaining a high-regioselectivity using the '048 patent's calcium-promoted γ -alumina catalyst, particularly in view of Wright '451's teaching that conversion increases with temperature. *Id.* at 11–12 (citing Ex. 2010 ¶¶ 123–127). Finally, Patent Owner submits—without explanation—“additional evidence of long-felt need and failure of others.” *Id.* at 12 (citing Ex. 2010 ¶¶ 128–132).

(2) *Analysis of Objective Evidence of Nonobviousness*

Having considered the arguments and evidence presented by the parties, we are not persuaded that the claimed reaction and calcination

temperature ranges are sufficiently critical to overcome the presumption of obviousness. As explained further below, we are not persuaded by Patent Owner's evidence of unexpected results, long-felt need, and failure of others.

(a) *Unexpected Results*

“To be particularly probative, evidence of unexpected results must establish that there is a difference between the results obtained and those of the closest prior art, and that the difference would not have been expected by one of ordinary skill in the art at the time of the invention.” *Bristol-Myers Squibb Co. v. Teva Pharms USA, Inc.*, 752 F.3d 967, 977 (Fed. Cir. 2014). Given that Wright '451 teaches or suggests each limitation of the substitute claims, we find—and the parties appear to agree—that Wright '451 is the closest prior art.¹² See Ex. 2006 ¶ 119 (“*Wright '451* is the closest prior art to the substitute claims, which does not appear to be in dispute.”) (citing Ex. 2007, 151:3–6).

To show unexpected results, Patent Owner asserts that the '048 Patent shows that a catalyst calcined at 440 °C provides unexpectedly better 1-octene regioselectivity and higher conversion than the same catalyst calcined at 500 °C (as taught by Wright '451). RMTA 19 (citing Ex. 1001, 26:10–27:9; Ex. 2006 ¶¶ 20, 123). Specifically, according to Patent Owner, the catalyst calcined at 440 °C provides a 1-octene regioselectivity of 96.7%

¹² Patent Owner asserts that a POSA would have found it unexpected that the '048 Patent's γ -alumina catalyst that is not silanized had increased conversion and regioselectivity compared to a γ -alumina catalyst that was silanized and then calcined at 400 °C. PO Reply RMTA 12 (citing Ex. 2010 ¶ 122). We agree with Petitioner, however, that that argument is not persuasive, as Patent Owner does not compare the results with Wright '451, which is the closest prior art. Pet. Sur-reply RMTA 3.

at a conversion of 97.2%, which is better than the catalyst calcined at 500 °C, which provides a regioselectivity of 93.1% at a conversion of 95.5%. *Id.* (citing Ex. 1001, 26:10–27:9).

We are not persuaded. “[A]ny superior property must be *unexpected* to be considered as evidence of non-obviousness.” *Pfizer, Inc. v. Apotex, Inc.*, 480 F.3d 1348, 1371 (Fed. Cir. 2007). We must, therefore, first consider what results were expected to determine what a POSA would have considered to be unexpected. *Id.* As seen by the table below, Wright ’451’s disclosure fully encompasses the temperature ranges and purported unexpected results recited by substitute claim 29 and shown in Table 2 of the ’048 Patent:

Reference	Reaction Temperature	Calcination Temperature	Conversion Rate¹³	Regioselectivity
Wright ’451	about 220 °C to about 440 °C	about 200 °C to 600 °C	greater than about 95%	about 90% to about 99%
Substitute Claim 29	240 °C to 360 °C	about 420 °C to about 480 °C	at least 95%	95% to about 99%
Table 2, Example 1	315 °C	440 °C	97.2%	96.7%
Table 2, Example 2	315 °C	500 °C	95.5%	93.1%

See Ex. 1039 ¶¶ 13, 67; RMTA A-1; Ex. 1001, 27:1–9.

¹³ In the table, we apply Patent Owner’s interpretation of “conversion” (i.e., the sum of the total percentage of alkenes) to address Patent Owner’s arguments. But we note that the conversion rates of Examples 1 and 2 of Table 2 under Petitioner’s interpretation would be 98.4% and 96.5%, respectively, which are also consistent with the conversion rate taught by Wright ’451. See Ex. 1001, 27:1–8 (Table 2).

In essence, Patent Owner argues that the claimed results and the results of Table 2 of the '048 Patent were unexpected even though those results are expressly taught by Wright '451. We cannot accept such a paradoxical argument. Wright '451 “is prior art for all that it teaches.” See *Beckman Instruments, Inc. v. LKB Produkter AB*, 892 F.2d 1547, 1551 (Fed. Cir. 1989). We, therefore, credit the testimony of Dr. Taber that because the results of Table 2 are consistent with the teachings of Wright '451, a POSA would not have considered those results to be unexpected. See Ex. 1046 ¶ 52 (citing Ex. 1039 ¶¶ 13, 14, 20, 22, 67, 78, 82).

Regarding the reaction temperature, Patent Owner points to the results in Table 5 and asserts that the decreased conversion of 93.3% at a reaction temperature of 370 °C is an unexpected result relative to the conversion of 96.9% at a reaction temperature of 350 °C. RMTA 20 (citing Ex. 2006 ¶¶ 126–127). According to Patent Owner, “*Wright '451* teaches that the lowest 1-butanol conversion is always at 340 °C and *conversion increases with temperature*” and “[n]othing in *Wright '451* would have taught or suggested to a POSITA that conversion of 1-hexanol would decrease at 370 °C.” *Id.* (citing Ex. 1039 ¶¶ 42–46; Ex. 2006 ¶¶ 125–127). Patent Owner asserts that the unexpected results are a difference in kind and not one of degree, “as even a 1–2% improvement in either conversion or 1-alkene regioselectivity can provide significant benefit from the perspective of reactor design and operating costs—for example, by eliminating a recycle stream.” RMTA 19–20 (citing Ex. 2006 ¶¶ 6, 20, 122–127).

We disagree. “[D]ifferences in degree’ of a known and expected property are not as persuasive in rebutting obviousness as differences in ‘kind’—i.e., a new property dissimilar to the known property.” *Bristol-Myers Squibb Co. v. Teva Pharms. USA, Inc.*, 752 F.3d 967, 977 (Fed. Cir.

2014); *see also* *UCB, Inc. v. Actavis Labs. UT, Inc.*, 65 F.4th 679, 693 (Fed. Cir. 2023) (stating “[a] difference of degree is not as persuasive as a difference in kind—i.e., if the range produces a new property dissimilar to the known property, rather than producing a predictable result but to an unexpected extent”) (internal quotations omitted).

Notwithstanding Dr. Wright’s conclusory testimony to the contrary, we find that a 2–3% improvement in conversion and regioselectivity is a difference in degree and not a “new property dissimilar to the known property” to qualify as a difference in kind. *Bristol-Myers Squibb*, 752 F.3d at 977. Dr. Wright fails to explain how the difference is a “difference in kind and not one of degree.” *See* Ex. 2006 ¶¶ 124, 127. To the extent Dr. Wright opines that the resulting benefit in “reactor design and operating costs” elevates the differences to differences “in kind,” we find *In re Harris*, 409 F.3d 1339 (Fed. Cir. 2005), to be instructive. In that case, the Federal Circuit found that a percentage increase in results was not a “difference in kind,” even where that difference resulted in “considerable cost savings.” *Id.* at 1344. Similarly, here, we are not persuaded by Dr. Wright’s testimony that the percentage increase in the known properties of conversion and regioselectivity amounts to a difference in kind, even with a “significant” benefit to “reactor design and operating costs.” *See* Ex. 2006 ¶¶ 124, 127.

Moreover, a showing of unexpected results must be commensurate in scope with the claimed range. *In re Harris*, 409 F.3d at 1344. We agree with Petitioner that Patent Owner’s asserted unexpected results are not commensurate in scope with substitute claim 29. Pet. Opp. RMTA 7–8; Pet. Sur-reply RMTA 2–3. Substitute claim 29 recites a calcination temperature range of “about 420 °C to about 480 °C” and a reaction temperature range of “240 °C to 360 °C.” RMTA A-1. Patent Owner asserts that the results from

a single calcination temperature at 440 °C (relative to calcination at 500 °C) and a single reaction temperature of 370 °C (relative to a reaction temperature of 350 °C) support the criticality of the claimed ranges. RMTA 19–20; PO Reply RMTA 11–12.

We find Patent Owner’s evidence is insufficient and consider the *E.I. DuPont* case to be analogous to the facts of this case. There, the patent owner attempted to show unexpected results by pointing to an experiment disclosed in the challenged patent’s specification. *E.I. DuPont*, 904 F.3d at 1011–12. The Federal Circuit was not persuaded, however, because the experiment tested only a single pressure that was not commensurate in scope with the claims, which recited a range of pressures. *Id.* at 1012. Similarly, here, we are not persuaded that a single data point is commensurate in scope with the recited calcination and reaction temperature ranges of the claims.

Alternatively, Patent Owner asserts that substitute dependent claims 34, 37, 40, and 43, which recite a narrower calcination temperature range of about 440 °C to about 460 °C, would be commensurate in scope if the calcination range of substitute claim 29 is too broad. RMTA 22–23 (citing Ex. 2006 ¶¶ 123–124, 130–131). Similarly, substitute dependent claims 35, 38, 41, and 44 recite a narrower reaction temperature range of 315 °C to 350 °C. RMTA 23 (citing Ex. 2006 ¶¶ 125–127, 132–133). Although these dependent substitute claims recite a narrower calcination range and reaction temperature range than substitute claim 29, Patent Owner’s argument nonetheless relies on a single data point (i.e., calcination at 400°C and reaction temperature of 370 °C) within the claimed range. Although Patent Owner’s counsel argued during oral argument that the narrower claims “are certainly commensurate in scope with the result at 440 degrees,” Tr. 41:17–24, that conclusory attorney argument does not sufficiently explain

how the alleged unexpected results apply to the full scope of the claimed temperature ranges. *See In re Peterson*, 315 F.3d at 1331 (finding unexpected results were not commensurate in scope with the claimed range of 1–3% where unexpected results were only shown with respect to 2%).

Having considered the arguments and evidence presented at trial, we are not persuaded that Patent Owner’s evidence shows the claimed reaction and calcination temperature ranges produced unexpected results over the teachings of Wright ’451.

(b) *Long-Felt Need and Failure of Others*

In a single sentence in its Reply, Patent Owner states that it “submits additional evidence of long-felt need and failure of others.” PO Reply RMTA 12 (citing Ex. 2010 ¶¶ 128–132). Patent Owner provides no analysis or explanation regarding its submitted evidence and instead cites five paragraphs of Dr. Wright’s Third Declaration. We agree with Petitioner that Patent Owner’s argument violates our rule against incorporation by reference. *See* 37 C.F.R. § 42.6(a)(3) (“Arguments must not be incorporated by reference from one document into another document.”); Pet. Sur-reply RMTA 5; Paper 27, 1. We, therefore, need not consider it.

Even if we were to consider Patent Owner’s arguments and evidence, we are not persuaded that Patent Owner has shown that a long-felt and unmet need or failure of others existed in light of Wright ’451’s teachings. That is, we are not persuaded by Patent Owner’s evidence given Wright ’451 teaches or suggests each limitation of the claims. *George M. Martin Co. v. Alliance Mach. Sys. Int’l LLC*, 618 F.3d 1294, 1304 (Fed. Cir. 2010) (“Where the differences between the prior art and the claimed invention are as minimal as they are here, . . . it cannot be said that any long-felt need was unsolved.”).

Nor are we persuaded that Patent Owner has satisfied its burden to show a nexus exists between the proposed substitute claims and the alleged long-felt need and failure of others. *Fox Factory, Inc. v. SRAM, LLC*, 944 F.3d 1366, 1373 (Fed. Cir. 2019) (“The patentee bears the burden of showing that a nexus exists.”). Dr. Wright asserts that “the ability of a catalyst to maintain regioselective dehydrogenation of 1-alcohols to 1-alkenes at high and steady conversion rates and for extended duration of time-on-stream, was and is a crucial consideration for any *commercial* and sustainable bio-1-alcohol dehydration process.” Ex. 2010 ¶ 130 (second emphasis added). The claims, however, do not require commercial-scale processes. Thus, we are not persuaded that a nexus exists between the alleged long-felt need and failure of others.

Having considered Patent Owner’s objective evidence of nonobviousness, we accord little weight to Patent Owner’s arguments regarding unexpected results, long-felt need, and failure of others. Accordingly, we find this case “involves a strong case of obviousness based on very close prior art and weak evidence of nonobviousness.” *E.I. DuPont*, 904 F.3d at 1013; *see also Newell Cos. v. Kenney Mfg. Co.*, 864 F.2d 757, 769 (Fed. Cir. 1988) (holding that objective evidence of nonobviousness may not be sufficient to outweigh strong evidence of obviousness).

3. *Conclusion on Obviousness of Substitute Claims 29–44*

Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that substitute claims 29–44 are unpatentable as obvious over Wright ’451.

E. Remaining Grounds

In the remaining challenges set forth in Petitioner’s Opposition to the RMTA, Petitioner asserts substitute claims 29–44 are unpatentable because

they fail to comply with the written description requirement, are indefinite, and are obvious over De Bruyn and Takamura. Pet. Opp. RMTA 2–5, 16–19. Petitioner also asserts that Patent Owner is estopped from adding the substitute claims under 37 C.F.R. § 42.73 and that the claims are procedurally improper. *Id.* at 19–25.

Because we have determined that substitute claims 29–44 are obvious over Wright ’451, which is dispositive as to all proposed substitute claims, we need not reach these remaining grounds. *See SAS Inst. Inc. v. Iancu*, 138 S. Ct. 1348, 1359 (2018) (holding that a petitioner “is entitled to a final written decision addressing all of the claims it has challenged”); *Bos. Sci. Scimed, Inc. v. Cook Grp. Inc.*, 809 F. App’x 984, 990 (Fed. Cir. 2020) (non-precedential) (recognizing that the “Board need not address issues that are not necessary to the resolution of the proceeding” and, thus, agreeing that the Board has “discretion to decline to decide additional instituted grounds once the petitioner has prevailed on all its challenged claims”).

III. CONCLUSION¹⁴

For the foregoing reasons, we conclude that Petitioner has shown by a preponderance of the evidence that proposed substitute claims 29–44 are unpatentable as obvious. We, therefore, grant-in-part Patent Owner’s

¹⁴ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

Revised Motion to Amend as to the noncontingent request to cancel claims 1–6, 8–10, and 22–28 of the '048 Patent and deny Patent Owner's Revised Motion to Amend as to the request to add substitute claims 29–44.

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
1–6, 8–10, 22, 23, 26– 28	103	Le ¹⁵		
24, 25	103	Le, Cheng ¹⁶		
1–6, 8–10, 22, 23, 26– 28	103	De Bruyn, Le ¹⁷		
24, 25	103	De Bruyn, Le, Cheng ¹⁸		
1–6, 8–10, 22, 23, 26– 28	103	Wright ¹⁹		
24, 25	103	Wright, Cheng ²⁰		
Overall Outcome				

Motion to Amend Outcome	Claims
Original Claims Cancelled by Amendment	1–6, 8–10, 22–28
Substitute Claims Proposed in the Amendment	29–44
Substitute Claims: Motion to Amend Granted	

¹⁵ Because these claims are canceled, we do not address this ground.

¹⁶ Because these claims are canceled, we do not address this ground.

¹⁷ Because these claims are canceled, we do not address this ground.

¹⁸ Because these claims are canceled, we do not address this ground.

¹⁹ Because these claims are canceled, we do not address this ground.

²⁰ Because these claims are canceled, we do not address this ground.

Substitute Claims: Motion to Amend Denied	29–44
Substitute Claims: Not Reached	

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Patent Owner’s Revised Motion to Amend is granted-in-part as to the noncontingent request to cancel claims 1–6, 8–10, and 22–28 of the ’048 Patent;

FURTHER ORDERED that Patent Owner’s Revised Motion to Amend is denied-in-part as to the request to add proposed substitute claims 29–44 to the ’048 Patent; and

FURTHER ORDERED that, because this is a Final Written Decision, any party to the proceeding seeking judicial review of this Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2023-01035
Patent 10,995,048 B2

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